

VIGNESH V

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Professional Summary

MCA graduate with a 6-month internship at the National Centre for Polar and Ocean Research (NCPOR), Ministry of Earth Sciences, Government of India, developing AI-powered search, chatbot, and data-management features for the National Polar Data Centre (NPDC). Hands-on experience in Python, Django, Retrieval-Augmented Generation (RAG), LLM APIs, and full-stack web applications, with additional experience in IoT and predictive analytics.

Technical Skills

Programming	Python, JavaScript, Java, C, C++
Web & Backend	Django, HTML, CSS, REST APIs
AI / ML	Machine Learning, NLP, scikit-learn
GenAI & Search	Retrieval-Augmented Generation (RAG), Fuzzy Matching, Hugging Face, Ollama, Groq/OpenRouter APIs
IoT & Data	ESP32, ThingSpeak, Sensor Integration, Data Analytics
Databases	SQLite, PostgreSQL
Tools	Git, GitHub, VS Code, Figma

Professional Experience

National Centre for Polar and Ocean Research (NCPOR) Dec 2025 – May 2026

Software Engineering Intern – Polar Data Service Section, Ministry of Earth Sciences, Government of India

- Architected a Retrieval-Augmented Generation (RAG) pipeline for an NLP-driven Intelligent Search and context-aware chatbot, retrieving relevant database records to generate summaries of scientific datasets.
- Designed a multi-provider LLM fallback architecture using Groq, OpenRouter, and local Ollama to maintain AI feature availability during provider failures, API limits, and network outages.
- Developed AI-assisted dataset submission workflows that automatically generated DIF-standard metadata, extracted geographic bounding coordinates, and mapped ISO topics, with role-based access control.
- Engineered a Smart Login system using session tracking and dual-table database lookups to provide contextual authentication feedback and identify unregistered users.
- Implemented Django caching and fuzzy matching to serve common queries locally, reducing unnecessary external API calls and improving response latency.

Projects

Indoor Air Quality Monitor with Predictive Analytics Jul 2025 – Nov 2025

ESP32, Python, ThingSpeak, scikit-learn

- Designed an IoT-based indoor air-quality monitoring system using an ESP32 microcontroller to capture real-time environmental data.
- Integrated automated data transmission with ThingSpeak for continuous cloud-based monitoring and visualization.
- Applied scikit-learn predictive analytics to historical sensor data to forecast upcoming air-quality trends.

Education

Pondicherry University 2024 – 2026

Master of Computer Applications (MCA)

Indira Gandhi College of Arts and Science 2021 – 2024

Bachelor of Computer Applications (BCA)

Trainings & Certifications

- **The Joy of Computing Using Python** – NPTEL (Jan 2025 – Apr 2025)
- **Data Analytics With Python** – NPTEL (Jan 2025 – Apr 2025)

Leadership & Activities

- **Class Representative (MCA):** Coordinated communication between students and faculty, supported academic activities, and organized department events.
- **Technical Paper Presentation:** Presented a paper on “IoT-based Air Quality Monitoring” at a university-level seminar.
- **Languages:** English (Fluent), Tamil (Native)